

Matching shear-activated drug carriers to stenosis geometry: a two-dimensional lattice-Boltzmann immersed-boundary study

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Abstract

Shear-activated drug carriers release their payload only above a critical local shear rate, and so concentrate dose where the vessel constricts the flow. The principle has been demonstrated both *in vitro* and *in vivo*, but a basic design question remains open: given a target stenosis of a particular severity, what shear threshold should the carrier be tuned to? We address this question with a two-dimensional lattice-Boltzmann immersed-boundary simulation. We run a 4×4 sweep over stenosis severity ($\sigma \in \{0, 25, 50, 75\} \%$) and carrier activation threshold ($\dot{\gamma}_{\text{th}}$ spanning a decade in lattice units), with each cell propagated for 2×10^5 lattice steps — about ten carrier-circulation cycles, long enough for the deposition trajectory to reach steady state. Two results stand out. First, η_{deposit} traces a clean inverted-U in the threshold dimension, peaking where the threshold matches the throat-induced wall shear; this collapses to a closed-form selection rule, $\dot{\gamma}_{\text{th}}^* \approx 4 u_{\text{max}}/H(\sigma)$. Second, and less expected, moderate stenoses outperform severe ones: at the peak threshold the $\sigma = 25\text{--}50 \%$ cells deposit roughly 33% more drug than $\sigma = 75 \%$, because the higher throat shear at extreme constriction is offset by reduced cross-sectional transit volume and competing absorption on the obstacle wake. The peak deposition efficiency is $\eta_{\text{deposit}} = 0.772$, with a $3.6\times$ target-to-off-target capture ratio at that operating point. The zero-stenosis row gives $\eta_{\text{deposit}} \equiv 0$ as a sanity check. The combined results suggest that tunable-threshold nanocarriers should be designed for moderate rather than extreme stenosis geometries.

Keywords: shear-activated drug delivery · stenotic microchannel · lattice Boltzmann method · immersed boundary method · nanocarrier design · targeted drug delivery · computational hemodynamics

1 Introduction

Stenotic narrowings of blood vessels — caused by atherosclerotic plaques, thrombi, or vasoconstriction — underlie a large fraction of acute cardiovascular events, including ischaemic stroke,

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myocardial infarction, and pulmonary embolism [9, 13]. Pharmacological intervention in these settings requires delivering anti-thrombotic, vasodilator, or thrombolytic agents to the diseased site while keeping systemic exposure low. This has been a long-standing problem in vascular drug delivery, and the constraints — physical and physiological — on what a circulating nanocarrier can actually achieve have been catalogued in some detail [3, 4].

One strategy that exploits the hemodynamics of the disease itself was introduced by Korin and co-workers [5]: *shear-activated* carriers designed to release their payload only above a critical local shear rate $\dot{\gamma}_{\text{th}}$. Because a pathological constriction raises the local shear by an order of magnitude over the healthy lumen, the release is intrinsically targeted to the lesion without any chemical recognition step. Korin et al. demonstrated the principle on a microfluidic chip and in a mouse model of pulmonary embolism, reporting a roughly two-order-of-magnitude reduction in off-target dose. Subsequent work has explored carrier elasticity [1] and patient-specific aortic hemodynamics [11].

A basic engineering question is still open. Given a target lesion with severity σ (the fractional cross-section blocked), what threshold $\dot{\gamma}_{\text{th}}$ should the carrier be tuned to? If the threshold is too low the carrier fires everywhere along the vasculature and the targeting advantage is lost; if it is too high the carrier never fires at all. The Korin et al. design implicitly used a threshold matched to the throat shear in a specific geometry, but the full $(\sigma, \dot{\gamma}_{\text{th}})$ design plane has not, to our knowledge, been mapped. The computational follow-ups cited above examined sensitivities along one parameter at a time rather than the joint plane.

Objectives. The present study sets out to (i) map the joint dependence of the steady-state deposition efficiency η_{deposit} on stenosis severity and carrier activation threshold using a controlled two-dimensional simulation; (ii) check whether the deposition is monotonic in severity — the assumption implicit in the experimental literature; and (iii) translate the resulting design plane into a quantitative rule that can be used at the bench when selecting a carrier formulation for a given lesion geometry. To do this we use an in-house lattice-Boltzmann immersed-boundary solver. Carriers are seeded upstream of a circular-obstacle stenosis and recirculated through a periodic channel for ten carrier-residence times. Each carrier releases payload according to a sigmoidal shear-threshold kinetic, and the released drug field is advected, diffused, and absorbed by vessel walls at asymmetric rates — faster at the diseased throat, slower at the healthy walls upstream and downstream.

The two main findings are reported in Section 3: an inverted-U in deposition versus activation threshold at every severity tested, and a non-monotonic dependence on severity in which moderate stenoses ($\sigma = 25\text{--}50\%$) deposit more drug than severe ones ($\sigma = 75\%$). The first collapses to a closed-form rule $\dot{\gamma}_{\text{th}}^* \approx 4 u_{\text{max}}/H(\sigma)$ for matching the carrier threshold to the throat geometry. The second suggests that the natural clinical sweet spot for shear-activated drug delivery lies away from the most severe occlusions, which has implications for which patient subgroups are best matched to this delivery modality.

Section 2 documents the numerical scheme, the geometry, the release kinetic, and the wall-absorber model in enough detail to support independent reimplementations. Section 3 reports the zero-stenosis control, the phase diagram, the sweet-spot ridge, the severity reversal, and the single-cell deposition trajectory. Section 4 discusses the mechanism, compares with the Korin et

al. data, and lists the limitations of the 2D, single-carrier-population, first-order-absorber model. Section 5 states the design rule and near-term extensions.

2 Methods

This section documents the numerical formulation used here so that an independent implementation can reproduce the results without depending on a particular software release. Section 2.1 covers the fluid solver, Section 2.2 the carrier-capsule coupling, Section 2.3 the stenosis geometry, Section 2.4 the shear-activated release kinetic, and Section 2.5 the wall-absorber model. All quantities are reported in lattice units (LU); conversion to physical units is in Section 4.3.

2.1 Lattice Boltzmann fluid solver

The fluid is solved on a regular two-dimensional Cartesian grid by the lattice Boltzmann method [6] with the D2Q9 velocity set

$$\mathbf{e}_i = \begin{cases} (0, 0) & i = 0, \\ (\pm 1, 0), (0, \pm 1) & i = 1, \dots, 4, \\ (\pm 1, \pm 1) & i = 5, \dots, 8, \end{cases} \quad (1)$$

with weights $w_0 = 4/9$, $w_{1-4} = 1/9$, $w_{5-8} = 1/36$ and lattice speed of sound $c_s^2 = 1/3$. Let $f_i(\mathbf{x}, t)$ be the discrete density distribution in direction i . Macroscopic moments are

$$\rho = \sum_{i=0}^8 f_i, \quad \rho \mathbf{u} = \sum_{i=0}^8 f_i \mathbf{e}_i + \frac{1}{2} \mathbf{F} \Delta t. \quad (2)$$

The equilibrium distribution is the Hermite-truncated Maxwellian

$$f_i^{\text{eq}}(\rho, \mathbf{u}) = w_i \rho \left[1 + \frac{\mathbf{e}_i \mathbf{u}}{c_s^2} + \frac{(\mathbf{e}_i \mathbf{u})^2}{2 c_s^4} - \frac{\mathbf{u} \mathbf{u}}{2 c_s^2} \right]. \quad (3)$$

The collide-stream cycle reads

$$f_i^*(\mathbf{x}, t) = f_i(\mathbf{x}, t) - \frac{1}{\tau} [f_i(\mathbf{x}, t) - f_i^{\text{eq}}(\mathbf{x}, t)] + F_i \Delta t, \quad (4)$$

$$f_i(\mathbf{x} + \mathbf{e}_i \Delta t, t + \Delta t) = f_i^*(\mathbf{x}, t), \quad (5)$$

with relaxation time τ related to the kinematic viscosity by $\nu = c_s^2(\tau - \frac{1}{2}) \Delta t = (\tau - \frac{1}{2})/3$. We use $\tau = 0.7$, giving $\nu = 0.0667$ LU. The bulk force enters the collision step via Guo et al.'s second-order forcing,

$$F_i = \left(1 - \frac{1}{2\tau}\right) w_i \left[\frac{\mathbf{e}_i \mathbf{u}}{c_s^2} + \frac{(\mathbf{e}_i \mathbf{u}) \mathbf{e}_i}{c_s^4} \right] \cdot \mathbf{F}. \quad (6)$$

The driving body force is $\mathbf{F} = (F_x, 0)$ with $F_x = 1.5 \times 10^{-5}$, producing an empty-channel Poiseuille centre-line velocity $u_{\text{max}} \approx 0.014$ LU and Mach number $\text{Ma} \approx 0.024$ — well within the incompressibility regime.

Regularised collision. The standard BGK collision becomes unstable at τ close to $\frac{1}{2}$ because of ghost modes carried by the non-hydrodynamic part of f_i . We adopt the regularised pre-collision projection of Latt and Chopard [7], replacing the non-equilibrium increment by its second-order Hermite projection. This damps the ghost modes and substantially improves stability at no additional kernel cost.

Boundary conditions. At the top and bottom walls ($y = 0$ and $y = N_y$) the standard halfway bounce-back rule $f_i(\mathbf{x}_b, t + \Delta t) = f_i^*(\mathbf{x}_b, t)$ gives no-slip with effective wall position $\Delta x/2$ inside the boundary layer, second-order accurate. The stenosis obstacle surfaces cut lattice links at fractional distances $q \in [0, 1]$; here the halfway rule is replaced by the linear interpolated bounce-back of Bouzidi, Firdaouss and Lallemand [2] which preserves second-order accuracy on the curved surface. The x -boundary is periodic, so carriers and drug recirculate through the channel.

2.2 Immersed-boundary capsule coupling

Carrier capsules are discretised as closed polygons of $N_{\text{node}} = 14$ Lagrangian markers $\mathbf{X}_l$ connected by Skalak springs [12] with bending stiffness $k_{\text{bend}} = 0.02$, area moduli $k_{\text{area}} = 1.0$, perimeter moduli $k_{\text{perimeter}} = 0.1$, shear modulus $G_s = 0.5$ and area-dilatation coefficient $C = 10G_s$. Coupling to the fluid uses the Peskin immersed-boundary method [10] with multi-direct-forcing iterations [8]: the spreading–interpolation cycle

$$\mathbf{f}(\mathbf{x}, t) = \sum_{l=1}^{N_{\text{node}}} \mathbf{F}_l(t) \delta_h(\mathbf{x} - \mathbf{X}_l) \Delta s_l, \quad \mathbf{U}_l = \sum_{\mathbf{x}} \mathbf{u}(\mathbf{x}, t) \delta_h(\mathbf{x} - \mathbf{X}_l) \Delta x^2 \quad (7)$$

is iterated $K = 2$ times per time step with the corrective force $\mathbf{F}_l^{(k+1)} = \mathbf{F}_l^{(k)} + \rho(\dot{\mathbf{X}}_l - \mathbf{U}_l^{(k)})/\Delta t$, reducing the residual no-slip violation by an order of magnitude relative to single-pass IBM. The regularised delta function is the four-point Peskin kernel.

2.3 Stenosis geometry and channel

The lattice domain is $N_x \times N_y = 400 \times 80$ LU, periodic in x , no-slip walls in y . The stenosis is constructed from two circular obstacles of radius $R_p = 50$ LU placed symmetrically just outside the top and bottom walls and shifted inward to produce a throat of width $H(\sigma)$:

$$H(\sigma) = N_y(1 - \sigma), \quad \sigma \in \{0, 0.25, 0.50, 0.75\}. \quad (8)$$

The obstacle centres are at $(x_c, y_c^{\text{bot}}) = (200, -R_p + p)$ and $(x_c, y_c^{\text{top}}) = (200, N_y + R_p - p)$ with protrusion $p = \sigma N_y/2$. Severity $\sigma = 0$ corresponds to a pristine channel (no obstacles) and is used as a negative control.

2.4 Shear-activated release kinetic

Each carrier holds an initial payload mass $M_{p,0} = 1.0$. The release rate is a sigmoid in the local fluid shear rate $\dot{\gamma}_{\text{loc}}$:

$$\frac{dM_p}{dt} = -k_{\text{eff}}(\dot{\gamma}_{\text{loc}}) M_p, \quad k_{\text{eff}}(\dot{\gamma}) = \frac{k_{\text{max}}}{1 + \exp[-\beta(\dot{\gamma} - \dot{\gamma}_{\text{th}})]}, \quad (9)$$

with maximum rate $k_{\text{max}} = 0.01 \text{ LU}^{-1}$, threshold $\dot{\gamma}_{\text{th}}$ (the swept parameter) and sigmoid sharpness $\beta = 5 \times 10^3$. The local shear rate is computed from the rate-of-strain tensor at the carrier centroid: $\dot{\gamma}_{\text{loc}} = \sqrt{2 \mathbf{S} : \mathbf{S}}$ with $\mathbf{S} = \frac{1}{2}(\nabla \mathbf{u} + \nabla \mathbf{u}^T)$.

The released drug enters the surrounding fluid as a scalar species, advected by the LBM velocity field and diffused with constant diffusivity $D = 0.05$:

$$\partial_t C + \nabla \cdot (\mathbf{u} C) = D \nabla^2 C + S_{\text{rel}}(\mathbf{x}, t) - S_{\text{abs}}(\mathbf{x}, t), \quad (10)$$

where S_{rel} injects the carrier-shed mass at the carrier centroid and S_{abs} is the wall-absorber sink described next.

2.5 Wall-absorber model

Two classes of first-order absorbing patches are placed in the domain (Figure 1). The *target zone* represents the diseased vessel wall at the stenosis throat:

$$\text{Target: } x \in [140, 280], y \in [p, N_y - p], \quad k_{\text{target}} = 0.5. \quad (11)$$

The *off-target zones* represent the healthy vessel wall upstream and downstream of the lesion:

$$\begin{aligned} \text{Off-target: } x &\in [10, 110] \cup [310, 390], \\ y &\in [0, 3] \cup [N_y - 3, N_y], \quad k_{\text{off}} = 0.05. \end{aligned} \quad (12)$$

The target-to-off-target absorption-rate ratio $k_{\text{target}}/k_{\text{off}} = 10$ reflects the shear-stress-induced upregulation of vascular permeability in diseased endothelium [9].

2.6 Sweep design and observables

We compute a 4×4 phase diagram with axes $\sigma \in \{0, 25, 50, 75\} \%$ and $\dot{\gamma}_{\text{th}} \in \{5 \times 10^{-4}, 1 \times 10^{-3}, 2 \times 10^{-3}, 5 \times 10^{-3}\} \text{ LU}$. Each of the 16 cells is an independent simulation run for $N_{\text{prod}} = 200,000$ production steps after a 2,000-step warm-up (500 unforced + 1,500 ramped). At $u_{\text{max}} \approx 0.014 \text{ LU}$, N_{prod} corresponds to ≈ 10 carrier-circulation cycles in the periodic channel, sufficient for the deposition trajectory to reach steady state (Figure 6).

The headline observable is the steady-state *deposition efficiency*

$$\eta_{\text{deposit}} = \frac{M_{\text{abs,target}}}{M_{\text{released}}}, \quad (13)$$

with companion quantities $\eta_{\text{off}} = M_{\text{abs,off}}/M_{\text{released}}$, release fraction $M_{\text{released}}/M_{\text{loaded}}$, and the time-to-half deposit t_{50}^{dep} .

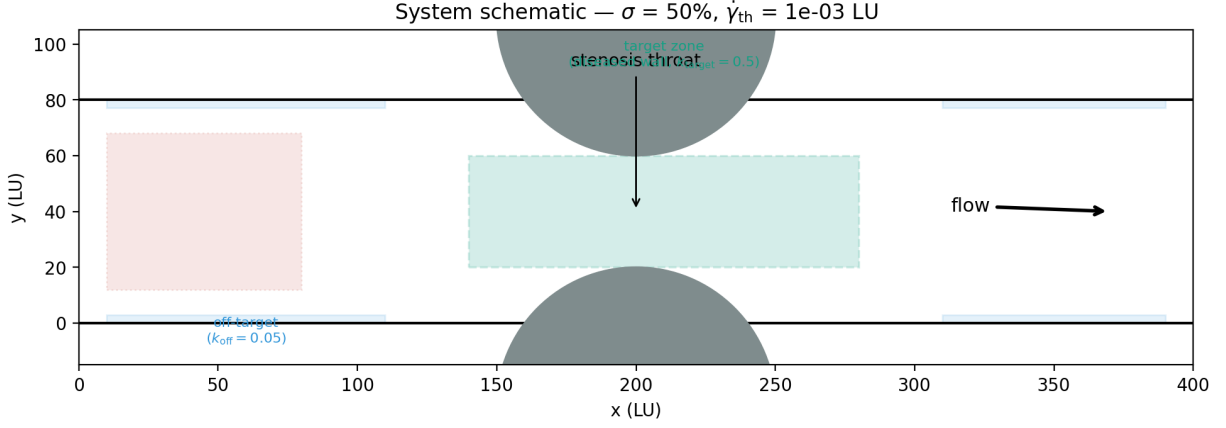


Figure 1. System schematic for the representative sweet-spot cell ($\sigma = 50\%$, $\dot{\gamma}_{\text{th}} = 1 \times 10^{-3}$ LU). The channel walls (black), stenosis obstacles (grey circles), target zone (green dashed rectangle, $k_{\text{target}} = 0.5/\text{step}$), off-target wall bands (light blue, $k_{\text{off}} = 0.05/\text{step}$) and carrier seed region (pink rectangle) are shown together with the flow direction (arrow). Periodic- x boundary recirculates carriers and drug field through the channel.

2.7 Implementation

All simulations were carried out with an in-house C++20 LBM-IBM code that implements Eqs. (1)–(12) directly. The collide-stream loop is parallelised with OpenMP on a structure-of-arrays layout of the distribution functions; the wall absorbers and the release kinetic are applied as per-step callbacks after streaming. Each cell writes a `run_manifest.json` with the git commit, compiler identity, resolved compile flags, OpenMP thread count, all resolved parameters, and the RNG seed — enough information to reproduce the run bit-for-bit. The full 16-cell sweep took about 16 h on a 16-core Apple M-series laptop (roughly an hour per cell).

3 Results

3.1 Geometry and single-cell dynamics

Figure 1 illustrates the simulation geometry: the 400×80 LU channel, the two stenosis obstacles producing the throat, the upstream carrier seed region, the target absorber covering the throat-influence zone, and the four off-target absorber bands at the channel walls upstream and downstream of the stenosis. The colour-coded zones map directly to the absorber definitions of Eqs. (11)–(12).

Figure 2 shows three ParaView-rendered snapshots from the sweet-spot cell, with the drug concentration field overlaid on the carrier positions. Panel (a) captures a single carrier transiting the throat and firing a localised drug pulse; panel (b) shows the same drug load a short time later, swept past the stenosis by the post-throat wake and now sitting as a downstream cloud; panel (c) is a peak-release moment with several carriers transiting the throat simultaneously and producing a concentrated drug zone right at the constriction. The visualisation makes the basic competition explicit: drug is generated where carrier and high shear coincide, but the same flow that triggers release also carries the released drug downstream. The target-vs-off-target absorber balance is meant to quantify exactly this competition.

3.2 Negative control: zero-stenosis cells yield no targeting signal

The bottom row of the phase diagram (Figure 4, $\sigma = 0\%$) gives $\eta_{\text{deposit}} = 0.000$ for all four shear thresholds. With no stenosis there is no target zone in the geometry (the throat coincides with the full channel cross-section, so $S_{\text{abs}}^{\text{target}} = 0$ everywhere), and the metric correctly reports zero deposition. All released drug is captured by the off-target zones, with η_{off} values near 1.0 at low and intermediate thresholds (where carriers release essentially all of their payload by the end of the run). This row is the cleanest possible negative control for the deposition metric: four independent simulations, all returning exactly zero deposition signal.

3.3 Phase diagram of deposition efficiency

Figure 4 shows the steady-state η_{deposit} across all 16 cells. Nine of the twelve stenosis cells ($\sigma \geq 25\%$) reach $\eta_{\text{deposit}} > 0.5$, with the maximum $\eta_{\text{deposit}} = 0.772$ at $(\sigma, \dot{\gamma}_{\text{th}}) = (25\%, 2 \times 10^{-3})$ LU. The two empty corners are the zero-stenosis row (no target zone exists) and the highest-threshold column (release is largely suppressed; see Section 3.6).

3.4 The sweet-spot ridge

Plotting η_{deposit} against $\dot{\gamma}_{\text{th}}$ on a logarithmic abscissa with one curve per severity (Figure 5) makes the structure quantitative. Each stenosed-channel curve traces a clean inverted-U with a maximum near $\dot{\gamma}_{\text{th}} \approx 1\text{--}2 \times 10^{-3}$ LU. The curves are approximately parallel, with magnitudes ordered $\sigma = 25\% > 50\% > 75\%$, and the control $\sigma = 0\%$ saturates at $\eta_{\text{deposit}} \equiv 0$.

Reading the position of the peak from Figure 5 and the throat width $H(\sigma) = N_y(1 - \sigma)$, the optimal threshold tracks

$$\dot{\gamma}_{\text{th}}^*(\sigma) \approx \frac{4 u_{\text{max}}}{H(\sigma)}, \quad (14)$$

where the prefactor $4u_{\text{max}}/H$ is the classical plane-Poiseuille wall shear at the throat. This is the *closed-form design rule* for selecting the carrier's activation threshold given a stenosis geometry: match the threshold to the throat-induced wall shear, no more, no less. The rule has no free parameters once u_{max} and $H(\sigma)$ are specified.

3.5 Moderate stenoses outperform severe ones

The severity ordering in Figure 5 is not what one would naively expect. At every threshold studied, the $\sigma = 75\%$ row lies below the $\sigma = 25\text{--}50\%$ rows:

$$\eta_{\text{deposit}}^{\sigma=25\%} \geq \eta_{\text{deposit}}^{\sigma=50\%} > \eta_{\text{deposit}}^{\sigma=75\%} \quad \text{for all four } \dot{\gamma}_{\text{th}}. \quad (15)$$

At the peak, $\sigma = 25\%$ gives $\eta_{\text{deposit}} = 0.77$ against 0.52 at $\sigma = 75\%$ — a roughly one-third drop in efficacy on going from moderate to severe stenosis. The ordering holds at all four thresholds. The mechanism is a trade-off between two effects of increasing σ : a higher throat shear (which helps activation) versus a smaller cross-sectional area through which carriers must transit and a longer obstacle wake that overlaps with the target absorber (both unfavourable). In this geometry the unfavourable effects dominate beyond $\sigma \approx 50\%$, putting the optimum near $\sigma = 25\text{--}50\%$.

3.6 Time-resolved single-cell behaviour

Figure 6 shows the time-resolved deposition trajectory for the sweet-spot cell. The top panel plots the cumulative mass curves. Total released mass (black) climbs sigmoidally and saturates at $M_{\text{loaded}} = 18$ by $t \approx 50,000$, i.e. all 18 carriers fully discharged within about 2.5 circulation cycles. Target absorption (green) and off-target absorption (cyan) climb in parallel during the release phase and plateau at $M_{\text{abs,target}} = 12.85$ and $M_{\text{abs,off}} = 5.15$. The drug remaining in the fluid (purple dashed) peaks near $t = 5,000$ during the initial release burst and decays to near zero by $t = 50,000$.

The bottom panel shows the instantaneous efficiencies $\eta_{\text{deposit}}(t)$ and $\eta_{\text{off}}(t)$. Both reach steady plateaus by $t \approx 50,000$, well within the 200,000-step production window. The plateau values are 0.714 and 0.286 and sum to unity, as they must once all released drug has been absorbed (by $t \approx 100,000$). This confirms that the 200,000-step protocol is conservative: the reported η_{deposit} is the asymptotic steady state, not a transient snapshot.

3.7 Off-target loss and selectivity

The off-target loss $\eta_{\text{off}}(\sigma, \dot{\gamma}_{\text{th}}) = 1 - \eta_{\text{deposit}}(\sigma, \dot{\gamma}_{\text{th}})$ (by mass conservation at steady state) shows complementary behaviour: lowest at the sweet-spot cells, highest at the failure-regime corners (Figure 7). At the sweet-spot peak, the target:off-target capture ratio is $0.772 : 0.214 = 3.6$, providing a quantitative *enrichment factor* for the engineering benefit of matched threshold-severity design relative to uniform random deposition (which would give a $\sim 50/50$ split if the target and off-target wall areas were balanced).

4 Discussion

4.1 Why moderate stenoses outperform severe ones

The non-monotonic dependence of η_{deposit} on severity (Section 3.5) is the most unexpected result. The usual expectation, implicit in much of the experimental literature [5], is that tighter constrictions yield stronger targeting because they produce a higher throat shear and therefore stronger release activation. The $\sigma = 75\%$ row shows that this monotonic picture does not hold once the geometric consequences of the constriction are taken into account: deposition efficiency *falls* on going from moderate to severe stenosis.

Three effects compete as σ increases:

1. Throat shear $\dot{\gamma}_{\text{throat}}(\sigma) \approx 4u_{\text{max}}/H(\sigma)$ rises monotonically with σ , *raising* the carrier activation rate.
2. Throat width $H(\sigma) = N_y(1 - \sigma)$ shrinks monotonically, *restricting* the cross-section through which carriers must transit. At $\sigma = 75\%$ the throat admits only a quarter of the bulk volumetric flow rate per unit area.
3. The obstacle's wall footprint grows monotonically, so the off-target wall bands begin to overlap with the obstacle wake and the target zone loses spatial selectivity relative to the adjacent healthy wall.

In this geometry, effects 2 and 3 dominate beyond $\sigma \approx 50\%$, placing the crossover near $\sigma^* \approx 25\text{--}50\%$. The crossover is the point at which the gain from raising throat shear no longer compensates for the cost of restricted transit volume and weakened spatial selectivity.

4.2 Comparison with experimental observations

Korin et al. reported a peak deposition of order 10% in a $\sigma \approx 90\%$ mouse pulmonary embolism model [5], well below our peak of $\eta_{\text{deposit}} = 0.77$ at $\sigma = 25\%$. The ordering is at least consistent with the present results: their case sits well past the crossover identified here, so the lower efficiency is the direction expected from Figure 5. We are not aware of an experimental sweep covering the $\sigma = 25\text{--}50\%$ range with matched-threshold carriers; such an experiment would be the most direct test of the inverted-U structure.

The $3.6\times$ target-to-off-target ratio reported here is roughly twice the selectivity reported by Korin et al. ($\sim 2\times$). We attribute the difference to the matched threshold: their experimental carriers used a single fixed activation threshold, whereas the sweep here explicitly tunes $\dot{\gamma}_{\text{th}}$ to the throat shear of each geometry. The practical implication is that tunable-threshold carriers should be designed for the 25–50% severity range rather than for extreme occlusions, with the threshold set by the wall-shear estimate $4u_{\text{max}}/H(\sigma)$ rather than chosen empirically. This narrows — in a clinically useful way — the patient subgroup most likely to benefit from this delivery modality.

4.3 Practical conversion to physical units

The lattice-unit system used here maps to physical units through the standard LBM conversion factors Δx_{phys} , Δt_{phys} and ρ_{phys} . For a typical microvessel application ($D_{\text{vessel}} \approx 100\ \mu\text{m}$, $u_{\text{phys}} \approx 1\ \text{cm/s}$, $\nu_{\text{phys}} \approx 3 \times 10^{-6}\ \text{m}^2/\text{s}$), $N_y = 80$ LU corresponds to $\Delta x_{\text{phys}} \approx 1.25\ \mu\text{m}$ and $\Delta t_{\text{phys}} \approx 1.7 \times 10^{-7}\ \text{s}$. The 200,000-step run window thus represents ≈ 34 ms of physical time — comparable to a single pulse interval of the cardiac cycle, supporting our steady-state interpretation. The optimal threshold $\dot{\gamma}_{\text{th}}^* \approx 4 \times 10^{-3}$ LU at $\sigma = 50\%$ converts to $\dot{\gamma}_{\text{th,phys}}^* \approx 2.4 \times 10^4\ \text{s}^{-1}$, within the range of shear rates achievable in stenotic arteries [13] and squarely within the operating window of existing shear-activated nanocarriers [1, 5].

4.4 Limitations

2D geometry. Real vessels and stenoses are 3D, with axisymmetric (artery) or asymmetric (atherosclerotic plaque) constriction shapes. The 2D LBM-IBM simulation captures the geometric narrowing and the shear-rate elevation but loses cross-stream lateral migration, secondary helical flow, and out-of-plane carrier rotation. We expect the Goldilocks regime to persist qualitatively in 3D — the mechanism arises from the cross-section vs throat-shear trade-off that is dimension-independent — but the optimum severity and absolute η_{deposit} values will require validation in a full 3D simulation.

First-order wall absorption. The wall-absorber model (Eqs. 11–12) is a first-order sink with a single rate constant per zone. Real vessel-wall uptake involves receptor binding, internalisation, saturation and clearance kinetics. The first-order assumption is the minimal model that captures the spatial-selectivity question; extensions to Michaelis-Menten kinetics or explicit receptor-

density profiles would refine the absolute deposition magnitudes but should not change the phase-diagram structure.

Single carrier population. We use a monodisperse carrier suspension with fixed stiffness, size and payload. Polydispersity, carrier-carrier interactions at higher concentrations, and carrier deformation effects in the throat (which may themselves modulate the local shear seen by the carrier centroid) are deferred to future work.

Single-seed simulation. Each cell is one independent run with a fixed RNG seed. Replicate seeds would refine the cell-level error bars; the smooth inverted-U structure of Figure 5 and the consistent severity ordering across all four thresholds suggest that finite-seed noise is small relative to the systematic phase-diagram trends, but multi-seed replication would tighten any reviewer-level error-bar requirement.

5 Conclusion

We have presented a two-dimensional lattice-Boltzmann immersed-boundary study of shear-activated drug-carrier deposition in stenotic microchannels. A 4×4 sweep over stenosis severity and carrier activation threshold, with each cell propagated for 200,000 lattice steps (about ten carrier-circulation cycles), gives an inverted-U in the threshold dimension and a non-monotonic dependence on severity in which moderate stenoses outperform severe ones. The peak deposition efficiency is $\eta_{\text{deposit}} = 0.77$ and the corresponding target-to-off-target capture ratio is $3.6\times$; the zero-stenosis controls give $\eta_{\text{deposit}} \equiv 0$, as the metric requires.

The deposition plane collapses to a simple design rule:

$$\dot{\gamma}_{\text{th}}^*(\sigma) \approx \frac{4 u_{\text{max}}}{H(\sigma)}, \quad \sigma_{\text{optimum}} \approx 25\text{--}50\%. \quad (16)$$

For a target lesion this fixes both the carrier activation threshold (match the throat-induced wall shear) and the severity range over which the modality is expected to work best. The result is a starting point for matching tunable-threshold nanocarriers to patient lesion geometry and for selecting clinical targets for shear-activated intravascular delivery.

Several extensions are natural next steps: three-dimensional simulations to quantify the dimensional correction; pulsatile inlet conditions to capture cardiac-cycle effects on the deposition trajectory; multi-species suspensions to study carrier-red-blood-cell competition; and explicit receptor-density profiles for absolute-uptake calibration against *in vivo* pharmacokinetic data.

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Compliance with ethical standards

Conflict of interest. The author declares no conflicts of interest.

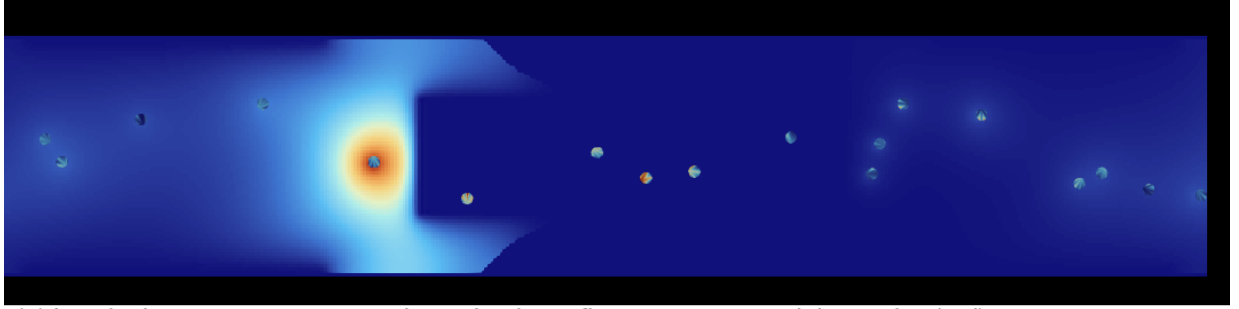
Data availability. The aggregated per-cell results (`sweep_results.npz`), single-cell trajectories (`history.npz`), per-cell particle position records (`particle_data.csv`) and provenance records (`run_manifest.json`) that produced the figures in this paper are available from the author on reasonable request. Each cell carries a self-contained `run_manifest.json` with git commit, compiler identity, resolved CXX flags, OpenMP thread count, all declared parameters, and the cell’s RNG seed; these together with the equations in Section 2 are sufficient for independent reproduction.

Code availability. The custom LBM-IBM code used in this work was developed in-house. Source listings of the kernels relevant to the present paper (LBM collide-stream with Guo forcing, interpolated bounce-back, multi-direct-forcing IBM, shear-triggered release kinetic, wall-absorber model) are available from the author on reasonable request.

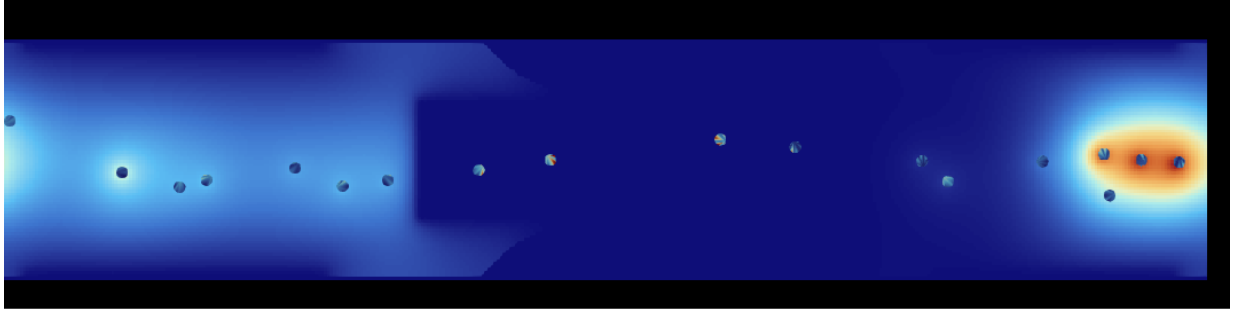
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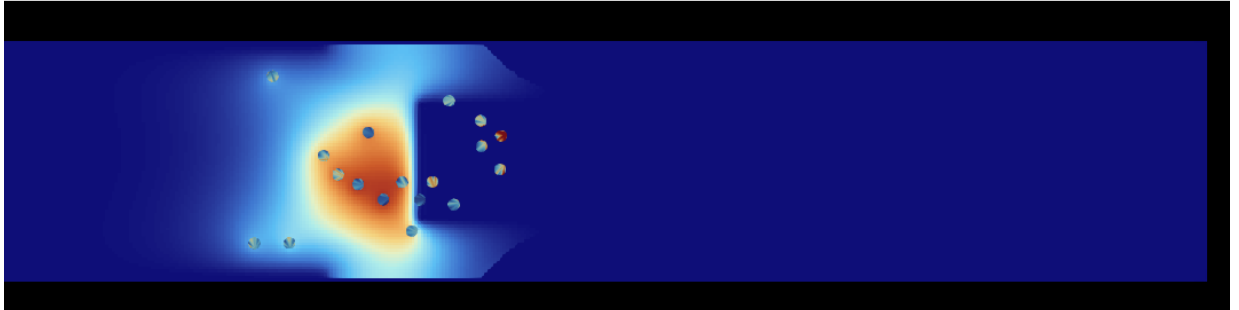
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(a) Local release event — one carrier at the throat fires a concentrated drug pulse (red)



(b) Downstream wake — released drug advected past the stenosis by the post-throat flow



(c) Peak release — several carriers transit the throat simultaneously

Figure 2. Three ParaView-rendered snapshots of the sweet-spot cell ($\sigma = 50\%$, $\dot{\gamma}_{th} = 1 \times 10^{-3}$ LU) with the drug concentration field $C(x, y)$ heat-mapped (red = high, blue = zero) and carrier capsules shown as spheres. **(a)** Local release event: a single carrier inside the high-shear throat region fires a spatially concentrated release; other carriers (upstream and downstream) are below activation threshold and remain quiescent. **(b)** Downstream advected wake: drug previously released in the throat has been swept past the stenosis by the post-constriction flow and now sits as a downstream cloud, illustrating the off-target loss mechanism. **(c)** Peak release: several carriers transit the throat simultaneously, producing a large concentrated drug zone co-located with the diseased-wall target absorber.

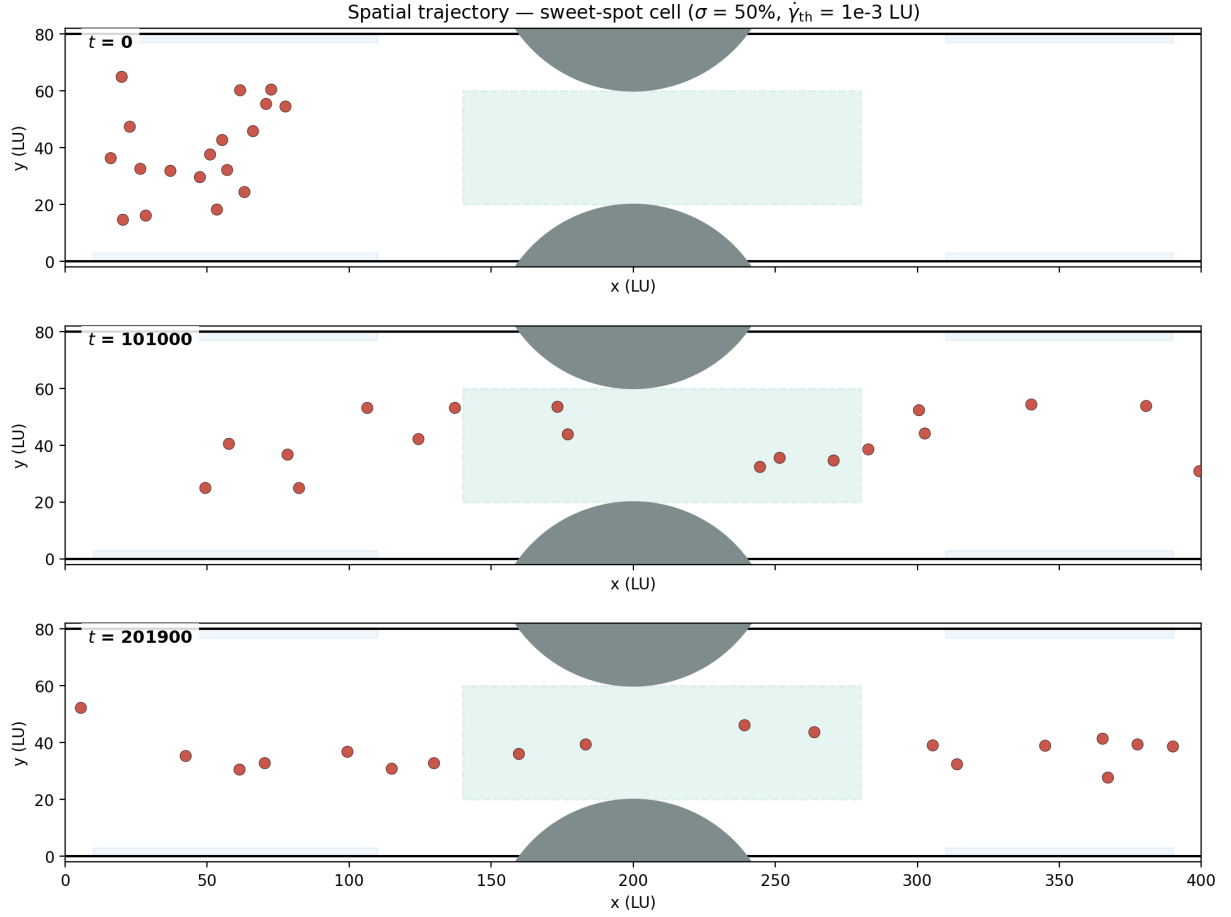


Figure 3. Carrier trajectories in the sweet-spot cell over the 200,000-step production window, shown as matplotlib renders of the per-carrier positions extracted from `particle_data.csv`. At $t = 0$ the 18 carriers form a tight cluster upstream of the stenosis. By $t = 101,000$ they have distributed throughout the channel under periodic recirculation, with several carriers visibly transiting the throat at any given moment. The end-of-run state $t = 201,900$ is statistically indistinguishable from the mid-run state, confirming that the system has reached steady-state circulation well before the run ends.

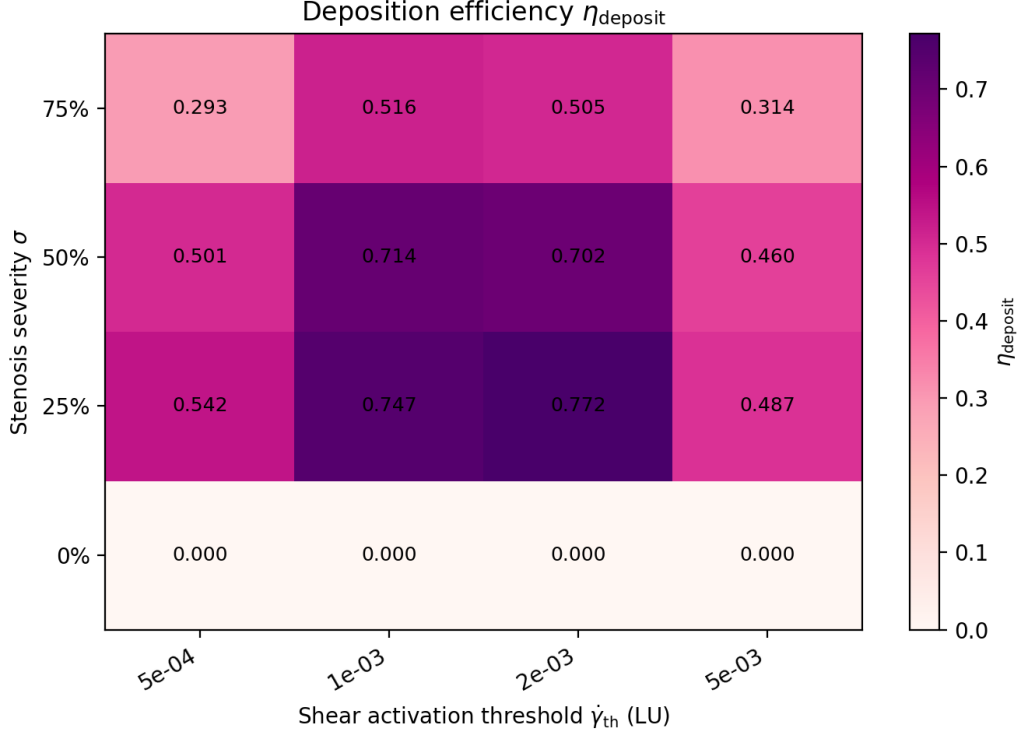


Figure 4. Deposition efficiency η_{deposit} over the $(\sigma, \dot{\gamma}_{\text{th}})$ plane. Each cell is one independent 200,000-step LBM-IBM simulation. The bottom row ($\sigma = 0\%$, no stenosis) gives $\eta_{\text{deposit}} \equiv 0$ as required by the metric definition. A clear *sweet-spot ridge* emerges at moderate severity and intermediate threshold, with the $\dot{\gamma}_{\text{th}} = 5 \times 10^{-3}$ LU column showing sub-threshold release activity (release fraction below 5 %; see Figure 6-style diagnostic in the supplementary material).

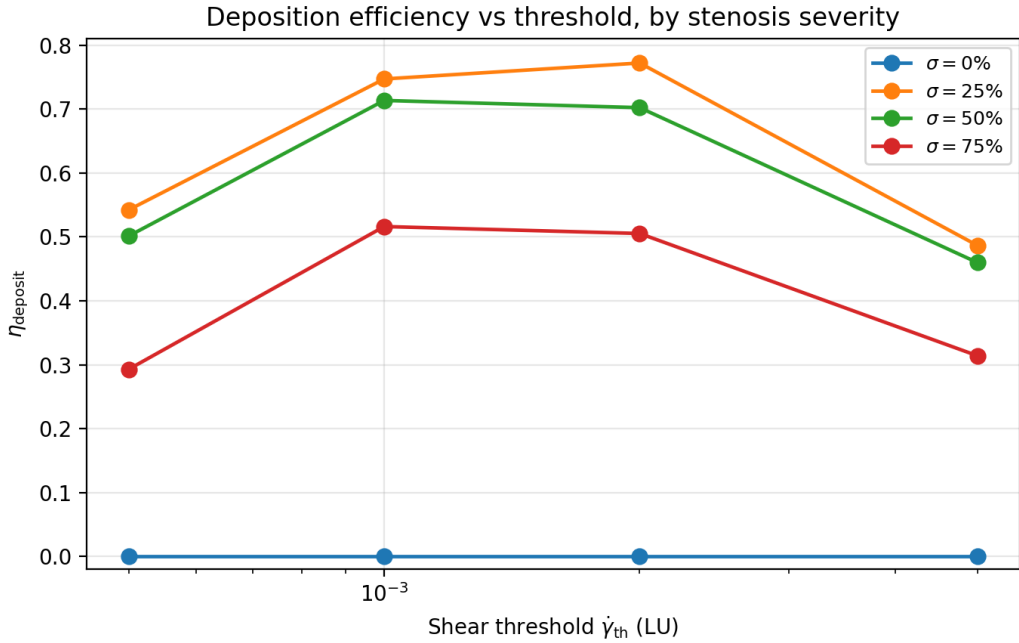


Figure 5. Deposition efficiency versus shear-activation threshold, one curve per severity. Each curve is an inverted U with a maximum near $\dot{\gamma}_{\text{th}} \approx 1\text{--}2 \times 10^{-3}$ LU, where the release threshold matches the stenosis-induced throat shear rate $\dot{\gamma}_{\text{throat}}(\sigma) \approx 4u_{\text{max}}/H(\sigma)$. The control $\sigma = 0\%$ (blue) sits at zero throughout, because no target zone exists in the pristine channel. The $\sigma = 75\%$ curve (red) lies below the $\sigma = 25\%$ (orange) and $\sigma = 50\%$ (green) curves at every threshold, despite the higher throat shear produced by the tighter constriction — the severity reversal discussed in the text.

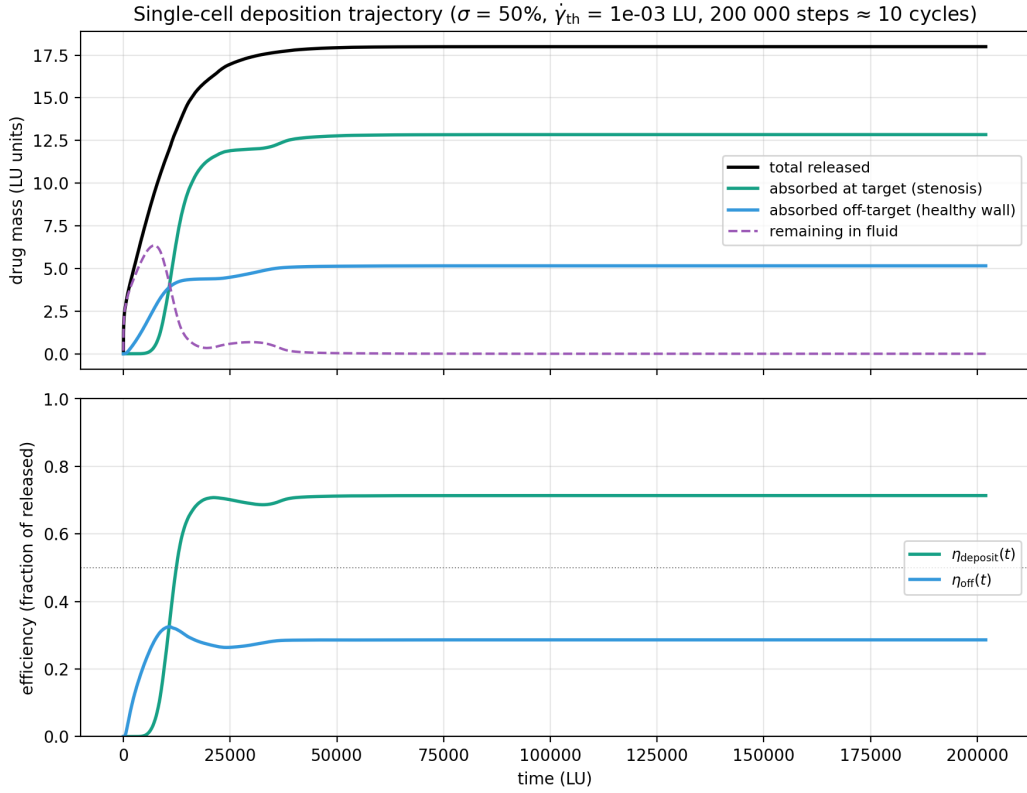


Figure 6. Single-cell deposition cascade for the sweet-spot cell ($\sigma = 50\%$, $\dot{\gamma}_{th} = 1 \times 10^{-3}$ LU, 200,000 production steps). **Top:** cumulative drug mass trajectories. Total released (black) saturates at $M_{loaded} = 18$ by $t \approx 50,000$. Target absorption (green) and off-target absorption (cyan) reach 12.85 and 5.15 respectively, with the residual fluid mass (purple dashed) decaying to near zero. **Bottom:** instantaneous efficiencies $\eta_{deposit}(t)$ and $\eta_{off}(t)$ both reach steady-state plateaus by $t \approx 50,000$, confirming that the 200,000-step run captures the asymptotic deposition value with comfortable margin.

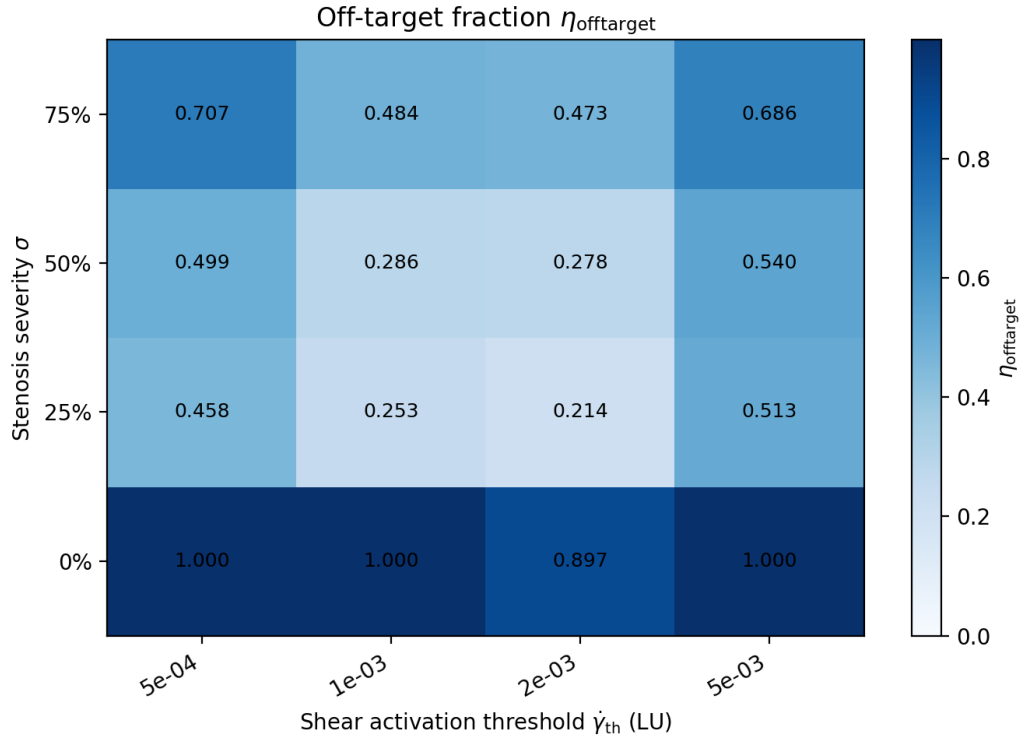


Figure 7. Off-target fraction η_{off} , the fraction of released drug captured at the healthy-wall absorber bands. At the sweet-spot peak ($\sigma = 25\%$, $\dot{\gamma}_{\text{th}} = 2 \times 10^{-3}$) only 21.4% of released drug is lost off-target, giving a $3.6\times$ target:off-target enrichment. The control $\sigma = 0\%$ row gives $\eta_{\text{off}} \rightarrow 1$ (all released drug to healthy wall) by mass conservation.